

2024-2025 春季学期天文与空间科学学院广义相对论期末

考试时间：2025 年 6 月 16 日

特别说明：考试分闭卷与开卷两部分，各一小时，闭卷做完可提前开始开卷部分，总分 100

考试给出曲率张量表达式 $R_{\lambda\mu\nu}^{\rho} = \Gamma_{\lambda\nu,\mu}^{\rho} - \Gamma_{\lambda\mu,\nu}^{\rho} + \Gamma_{\sigma\mu}^{\rho}\Gamma_{\lambda\nu}^{\sigma} - \Gamma_{\sigma\nu}^{\rho}\Gamma_{\lambda\mu}^{\sigma}$

闭卷部分

一、概念与简答

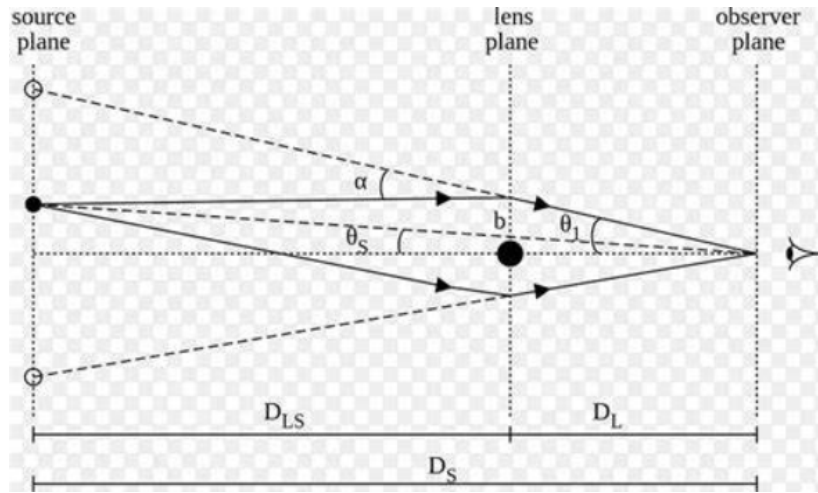
1. 伯克霍夫定律
2. 测地线方程及其意义
3. 等效原理
4. 谐和坐标及其意义
5. RW 度规
6. 史瓦西黑洞内外时空性质比较
7. 用张量方程性质证明 Bianchi 恒等式，并证明爱因斯坦张量 $G^{\mu\nu}$ 协变微分为 0
8. 写出质量为 M 球壳的真空内外度规解

开卷部分

二、计算题

- 给定抛物面空间部分度规 $dl^2 = (1 + 4r^2)dr^2 + r^2d\phi^2$ ，其中 ϕ 就为一般定义的天顶角
 - 求所有度规分量
 - 计算所有 Christoffel 符
 - 判断 $\phi = \text{常数}$ 是否为测地线，为什么？
- 如图，参数 D_L, D_S, D_{LS} ，黑洞质量为 M ，黑洞正后方有一星体，因引力透镜效应导致爱因斯坦环，证明角半径为

$$\theta_E = \sqrt{\frac{4MD_S}{D_{LS}D_L}}$$



(考试的时候这张图黑点表示的星体就在两个像正中)

- 基于下面关于 LIGO 在 GW150914 事件中探测到引力波的论文片段，回答问题：

The basic features of GW150914 point to it being produced by the coalescence of two black holes—i.e., their orbital inspiral and merger, and subsequent final black hole ringdown. Over 0.2 s, the signal increases in frequency and amplitude in about 8 cycles from 35 to 150 Hz, where the amplitude reaches a maximum. The most plausible explanation for this evolution is the inspiral of two orbiting masses, m_1 and m_2 , due to gravitational-wave emission. At the lower frequencies, such evolution is characterized by the chirp mass [11]

$$\mathcal{M} = \frac{(m_1 m_2)^{3/5}}{(m_1 + m_2)^{1/5}} = \frac{c^3}{G} \left[\frac{5}{96} \pi^{-8/3} f^{-11/3} \dot{f} \right]^{3/5},$$

where f and $\dot{f}$ are the observed frequency and its time derivative and G and c are the gravitational constant and speed of light. Estimating f and $\dot{f}$ from the data in Fig. 1, we obtain a chirp mass of $\mathcal{M} \approx 30M_\odot$, implying that the total mass $M = m_1 + m_2$ is $\gtrsim 70M_\odot$ in the detector frame.

证明文中给出的公式，并讨论黑洞质量的相关性质